MACHINE LEARNING AND DEEP LEARNING

Lab 04: Self-Supervised Learning (SSL)

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SELF-SUPERVISED LEARNING (SSL)

- **Self-Supervised Learning (SSL)** is a Deep Learning technique that involves **pre-training** a deep model on some **pretext tasks.**
- This allows the model to learn **significant patterns** (and relevant features) in data before being applied to the main task.
- This technique is particularly powerful when the amount of labeled data available for training is **limited.**

ROTATION-BASED PRETEXT TASKS

- Pretext tasks can be anything pushing the model to learn some features useful to capture the meaning of the samples before the main task.
- A typical pretext task is **rotation:** each image in the training dataset is **randomly rotated** by some angle (typically 0, 90, 180, 270 deg).
- The goal of the model is to be able to **predict the rotation angle.**
- This forces to model to **learn useful representations** of each class before being actually used in a classification task.

CONTRASTIVE SSL

- Another branch of pretext tasks is **Contrastive Self-Supervised Learning.**
- The idea is to force the model to learn how to map two slightly different instances of the same image **close together** in the feature space, while mapping instances of different images **far apart.**
- This technique is called "contrastive" as the model learns how to extract similar features from **positive pairs** and different feature representations from **negative pairs.**
- Positive pairs are typically created using **two slightly different views** of the same image obtained through **random augmentations** applied to the same anchor image (random flipping, random grayscale, …).

EXERCISES

- The goal is to implement **two image classification tasks**, using **Self-Supervised Learning** to boost accuracy of models when training samples are limited.
- **Task 1:**
  - You will implement an **image classification task** on the MNIST dataset (digits).
  - You will train and compare two models, both trained using **few training samples.** One model will be trained entirely from scratch, the other will be **pre-trained** on a **rotation-based** pretext task.
- **Task 2:**
  - You will implement an **image classification task** on the CIFAR-10 dataset.
  - You will train and compare two models, both trained using **few training samples.** One model will be trained entirely from scratch, the other will be pre-trained using a **Contrastive Learning approach**.
- **Assignment:** complete *Self_supervised_learning.ipynb* following the instructions provided in the comments.